

Geometric–Topological Representations for Self-Reconfiguration Planning of Continuum Modular Robots

Zhefeng Huang¹, Yilin Cai¹, Yifan Wang¹ and Yue Chen²

Abstract—Modular self-reconfigurable robots adapt by reconnecting modules into new morphologies. Although graph-based representations are widely used, connectivity alone may not fully specify a configuration. We study a modular self-reconfigurable continuum robot and introduce a unified geometric–topological representation that captures module- and gripper-level connectivity, bending angles, an angle system encoding cyclic order and grasping angles and the outer-face boundary permutation. This representation supports formal definitions of shape and kinematic equivalence and enables self-reconfiguration to be formulated as task and motion planning over a locally infinite graph. In this graph, nodes represent self-collision-free deformation manifolds defined by fixed connectivity and angle-system constraints, while edges represent discrete grasping and releasing actions. Correspondence-based action discretization with our Hierarchical Expansion-As-Rollout Tree Monte Carlo Tree Search (HEART-MCTS) perform task-level search over inter-manifold transitions, while Atlas-RRT* computes collision-free continuous paths within individual deformation manifolds. Experiments on procedurally generated configurations and hardware demonstrations validate efficient planning between diverse robot morphologies.

I. INTRODUCTION

Modular self-reconfigurable robots (MSRRs) adapt their morphology by reconnecting repeated modules for different tasks and environments [1], [2]. In this work, we study the previously developed Modular Self-Reconfigurable Continuum Robot (MSRCR), designed for general-purpose locomanipulation [3], [4]. Each module combines a compliant continuum body with two grippers that serve as both manipulators and docking interfaces. This grasping-as-docking design enables chain and branched morphologies with quasi-freeform connection angles, as illustrated in Fig. 1. However, this quasi-freeform geometry introduces a representation challenge. Existing planners for chain-type MSRRs commonly represent configurations as connectivity graphs and use graph isomorphisms to establish correspondences between initial and target configurations, thereby reducing the planning search space [5], [6], [7], [8], [9]. But these graph-based formulations do not directly apply to MSRCR,

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¹Zhefeng Huang, Yilin Cai and Yifan Wang are with the Department of Mechanical Engineering, Georgia Institute of Technology, Atlanta, GA 30332 USA (e-mail: zhuang480@gatech.edu; yilincail@gatech.edu; wangyf@gatech.edu)

²Yue Chen is with the Department of Biomedical Engineering, Georgia Institute of Technology/Emory University, Atlanta, GA 30332 USA (e-mail: yue.chen@bme.gatech.edu)

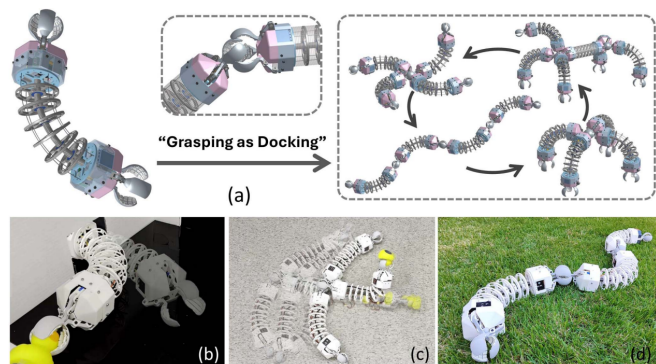


Fig. 1. The previously developed MSRCR [3], [4]. (a) Grasping-based docking enables transitions between morphologies. (b–d) Representative manipulation and locomotion behaviors.

whose continuous connection geometry cannot be captured by connectivity alone.

In this work, we focus on planar self-reconfiguration, where the MSRCR remains on an orientable flat support surface and stays connected and self-collision-free throughout the process. This setting avoids the need to model gravity-induced stability while retaining the central geometric and kinematic challenges of reconfiguration. Under the planar assumption, a configuration is specified not only by module connectivity, but also by continuous connection geometry, the planar embedding of the resulting graph, and ordered grasping relations among docked grippers. These properties motivate a representation that augments topology with the geometry of the embedded configuration.

Thus, we introduce a unified geometric–topological representation that combines module- and gripper-level connectivity, continuous bending and grasping angles, cyclic order, and the outer-face boundary permutation. Based on this representation, two equivalence relations—shape and kinematic equivalence—are formalized for comparing MSRCR configurations, and the reconfiguration state space is modeled as a locally infinite graph whose nodes are deformation manifolds and whose edges correspond to discrete grasping and releasing actions. Self-reconfiguration is then formulated as task-and-motion planning on this graph, with task planning over inter-manifold transitions and motion planning within individual manifolds. For task planning, HEART-MCTS, a Monte Carlo tree search over a Hierarchical Expansion-As-Rollout Tree, is developed, while Atlas-RRT* computes collision-free intra-manifold paths [10], [11]. This framework is evaluated on procedurally generated configuration pairs and demonstrated on the physical MSRCR.

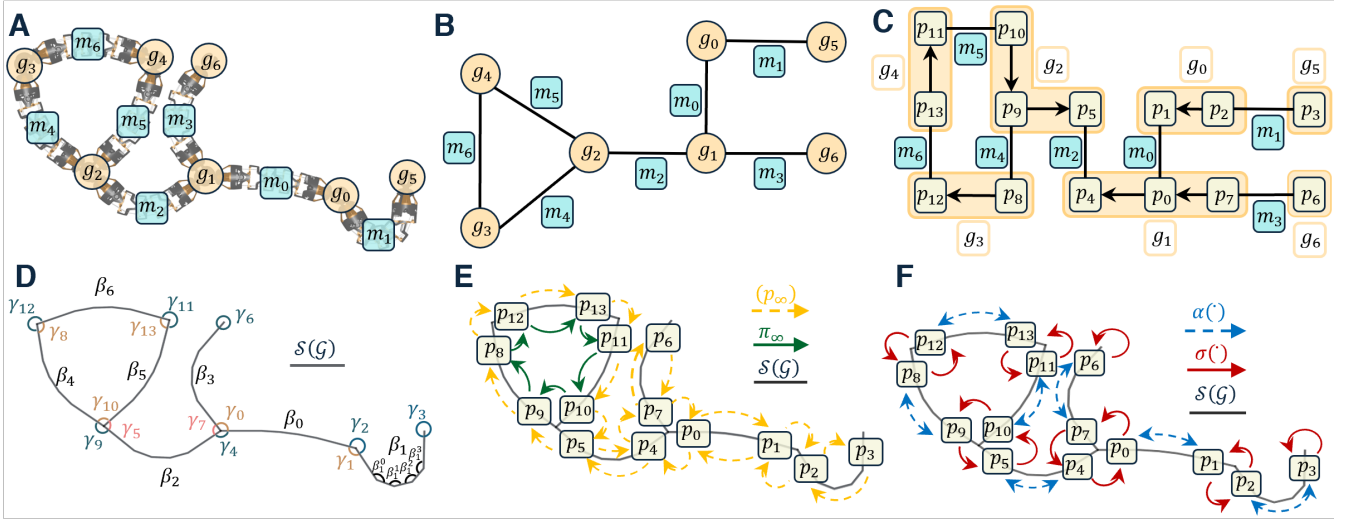


Fig. 2. MSRCR configuration representation. (A) Example planar configuration with seven modules and seven grippers; each gripper g is either a chain of grippers connected through pairwise grasping relations (from g_0 to g_4) or a single gripper not participating in any grasp (g_5 and g_6). (B) Module-level topology represented by the undirected graph $\mathcal{G}$. (C) Gripper-level connectivity structure represented by the mixed graph $\mathcal{G}_M$, with undirected edges for module links and directed edges for grasping relations. (D–F) The shape $\mathcal{S}$ of a configuration is given by a planar embedding of $\mathcal{G}$, whose edges are drawn as equiangular polylines. The bending angle β of a module is defined by its incident U-joint segments, and the grasping angle γ is defined between neighboring grippers within a grip. Interpreting each gripper p_i as a dart (half-edge) yields a combinatorial map R that encodes the cyclic ordering σ and head–tail gripper pairing α , together with an outer-face boundary permutation π_∞ obtained by traversing the outer boundary of the embedding clockwise.

II. GEOMETRIC–TOPOLOGICAL REPRESENTATION

A. Connectivity and Grasping Structure

We consider a planar MSRCR configuration with N_m continuum modules and N_g grippers, embedded on an oriented, flat, and effectively unbounded support plane $\mathbb{R}^2$. Each module m_i has a head and a tail gripper, denoted by p_{2i} and p_{2i+1} , respectively. A grip g_i consists either of multiple grippers docked together or a single undocked gripper. An example planar configuration is shown in Fig. 2A.

Formally, the modules M , grippers P , grasping order poset P_S , and grippers G are represented by

$$\begin{aligned} M &= \{m_i\}_{i=0}^{N_m-1}, & P &= \{p_i\}_{i=0}^{2N_m-1}, \\ P_S &= (P, \leq_S), & G &= \{g_i\}_{i=0}^{N_g-1}. \end{aligned} \quad (1)$$

Here, the partial order $p_i \leq_S p_j$ indicates that gripper p_i is enclosed within the grasp of p_j . Each grip $g_i \in G$ is defined as the ground set of a maximal chain in P_S , including singleton chains corresponding to undocked grippers. These elements induce two graph representations at two levels of abstraction. The module-level topology $\mathcal{G}$ is an undirected graph whose vertices are grippers and whose edges are modules (Fig. 2B). The gripper-level connectivity structure $\mathcal{G}_M$ is a mixed graph whose vertices are grippers, with undirected edges pairing the head and tail grippers of each module and directed edges encoding grasping relations (Fig. 2C). Thus, $\mathcal{G}$ captures module-level connectivity, whereas $\mathcal{G}_M$ preserves gripper-level structure and ordering of grasping relations.

B. Embedded Geometry

Having defined the connectivity structure, we next specify its planar geometric realization. The shape $\mathcal{S}(\mathcal{G})$ of a configuration is a planar embedding of $\mathcal{G}$, with each module

edge drawn as an equiangular polyline consistent with its bending state. The continuous geometry and local embedding structure are encoded by

$$\begin{aligned} B &= [\beta_0, \beta_1, \dots, \beta_{N_m-1}]^T, \\ \Gamma &= [\gamma_0, \gamma_1, \dots, \gamma_{2N_m-1}]^T, \\ R &= (P, \sigma, \alpha), & A &= (R, \Gamma). \end{aligned} \quad (2)$$

Here, the bending angle β_i parameterizes module m_i , while γ_i denotes the grasping angle from gripper p_i to its counter-clockwise successor within the same grip (Fig. 2D). Interpreting each gripper as a dart yields the combinatorial map R , where σ encodes the cyclic order of grippers within each grip and the involution α pairs the head and tail grippers of each module (Fig. 2F). In analogy to the rotation system encoded by R , we define an angle system $A = (R, \Gamma)$ by augmenting the combinatorial map with continuous grasping angles Γ .

Local cyclic order does not, by itself, identify the unbounded face of an embedding in $\mathbb{R}^2$. We therefore traverse the outer boundary clockwise to obtain the gripper sequence (p_∞) , shown by the dashed arrows in Fig. 2E. After removing repeated entries, the outer-face boundary permutation π_∞ maps each gripper in P_∞ to its successor in this traversal. The complete configuration representation and its induced planar shape embedding are then written as

$$\begin{aligned} \mathcal{C} &= \{M, P_S, \mathcal{G}, B, A, \pi_\infty\}, \\ \mathcal{S} &: (\mathcal{G} \mid B, A, \pi_\infty) \hookrightarrow \mathbb{R}^2. \end{aligned} \quad (3)$$

Together, these components specify both which robot elements are connected and how the resulting structure is geometrically embedded in the plane.

C. Shape and Kinematic Equivalence Classes

We formalize two equivalence relations for MSRCR configurations. First, two configurations $\mathcal{C}_0$ and $\mathcal{C}_1$ are shape equivalent, denoted by $\mathcal{C}_0 \sim_s \mathcal{C}_1$, if there exists a planar rigid-body transformation $T \in SE(2)$ such that $T(\mathcal{S}_0(\mathcal{G}_0)) = \mathcal{S}_1(\mathcal{G}_1)$. Shape equivalence therefore provides the criterion for determining whether a resulting configuration matches the target shape, independent of its global planar pose.

For a configuration $\mathcal{C}_0$ with bending-angle vector B_0 , a deformation is defined by a continuous trajectory

$$\tau : [0, 1] \rightarrow \mathbb{R}^{N_m}, \quad \tau(0) = B_0, \quad (4)$$

which produces a time-varying configuration $\mathcal{C}(t)$ with shape embedding $\mathcal{S}(t)$. A deformation is called valid if $\mathcal{S}(t)$ remains self-collision-free and $A(t) = A(0)$ for all $t \in [0, 1]$. Therefore, a valid deformation changes only the module bending angles while preserving connectivity and docking angles. The resulting continuous family of shape embeddings $\{\mathcal{S}(t)\}_{t \in [0, 1]}$ induces an angle-system-preserving isotopy in $\mathbb{R}^2$.

Two configurations $\mathcal{C}_0$ and $\mathcal{C}_1$ are kinematically equivalent, denoted by $\mathcal{C}_0 \sim_k \mathcal{C}_1$, if a valid deformation connects $\mathcal{C}_0$ to a configuration $\mathcal{C}'_0$ that is shape equivalent to $\mathcal{C}_1$, i.e., $\mathcal{C}'_0 \sim_s \mathcal{C}_1$. Kinematic equivalence therefore determines whether one configuration can be transformed, solely through valid deformation, into a shape-equivalent realization of another. Correspondingly, configurations kinematically equivalent to $\mathcal{C}_0$ constitute a kinematic-equivalence class $[\mathcal{C}_0]_k$.

Direct verification of the kinematic equivalence defined above requires establishing the existence of a physically realizable bending trajectory. As a necessary preliminary test, we instead check for an angle-system-preserving isotopy, which exists if and only if the configurations have matching angle systems and outer-face boundary permutations under an appropriate gripper relabeling. This condition is used during task planning to rule out configurations that cannot be kinematically equivalent. Candidate transitions are then screened through constrained endpoint optimization, while physical reachability is verified through motion planning. These procedures are detailed later.

III. SELF-RECONFIGURATION PLANNING

A. Problem Formulation

Given an initial configuration $\mathcal{C}_0$ and a target configuration $\mathcal{C}_1$, self-reconfiguration planning seeks a finite sequence of valid actions $\mathcal{A} = (a_0, a_1, \dots, a_T)$ that transforms $\mathcal{C}_0$ into a configuration $\mathcal{C}'_0$ that is shape equivalent to $\mathcal{C}_1$ under an appropriate gripper correspondence $f_c : P_0 \rightarrow P_1$, denoted by $\mathcal{C}'_0 \sim_{s|f_c} \mathcal{C}_1$. The correspondence f_c is a bijection that preserves the head-tail pairing of each module, i.e., $f_c \circ \alpha_0 = \alpha_1 \circ f_c$, and thereby induces a consistent correspondence between modules. Three action primitives are considered: a releasing action $a_r(p)$ removes an existing grasping relation associated with gripper p ; a deforming action $a_d(\tau)$ applies a valid bending-angle trajectory τ ; and a grasping action $a_g(\pi, s, \gamma)$ closes a path $\pi = (p_{i_0}, p_{i_1}, \dots, p_{i_k})$ in $\mathcal{G}_M$ by making p_{i_0} grasp p_{i_k} at angle γ . The resulting cycle

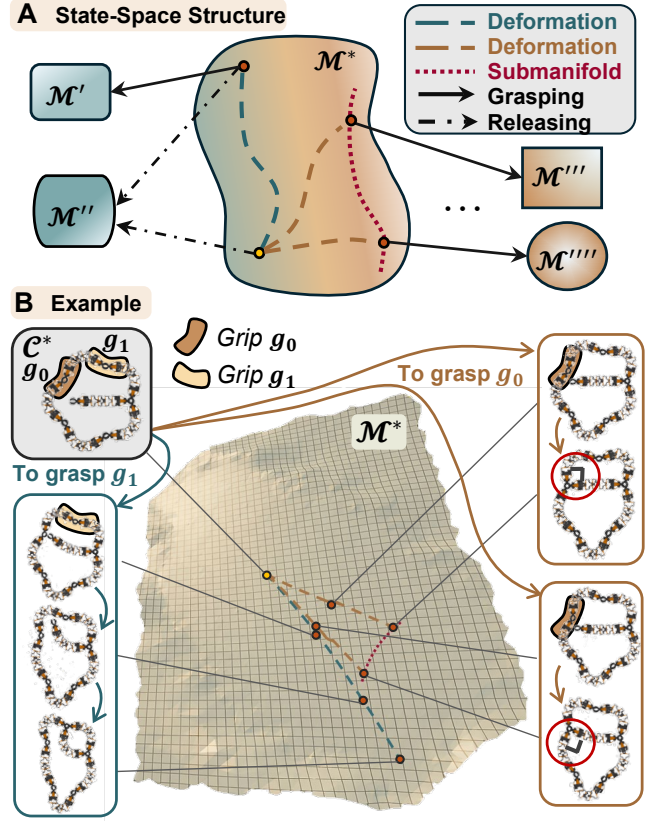


Fig. 3. State-space formulation for MSRCR self-reconfiguration. (A) The state space is represented as a locally infinite graph whose nodes are deformation manifolds and whose edges correspond to grasping and releasing actions. Deformations are continuous trajectories within a manifold, while grasping is enabled on submanifolds where a gripper overlaps its target grip. Varying the grasping angle connects such a submanifold to infinitely many neighboring manifolds. (B) Example grasp-preparation motions for a six-module configuration. Dashed trajectories on a sampled deformation manifold connect the initial configuration $\mathcal{C}^*$ to configurations that enable grasping grip g_0 or g_1 .

is oriented counterclockwise for $s = 1$ and clockwise for $s = -1$. Throughout the process, the robot must remain connected and self-collision-free, and each grasping angle remains fixed until the corresponding grasp is released.

For a configuration $\mathcal{C}^*$, its kinematic-equivalence class under identity correspondence forms a deformation manifold $\mathcal{M}^* \subset \mathbb{R}^{N_m}$, parameterized by module bending angles. Each point on $\mathcal{M}^*$ represents a valid configuration, and each continuous path within it represents a deformation. Grasping and releasing actions edit the connectivity structure and induce transitions between manifolds, while a grasp is enabled only on a submanifold where the relevant gripper and grip overlap. Because the grasping angle varies continuously, such a submanifold may connect to infinitely many neighboring manifolds, yielding a locally infinite graph whose nodes are deformation manifolds and whose edges are grasping and releasing actions, as illustrated in Fig. 3A. Self-reconfiguration is therefore formulated as hybrid path planning on this graph: task planning finds a discrete path of inter-manifold transitions, while motion planning computes continuous intra-manifold paths. Fig. 3B shows multiple such paths leading to different grasp-preparatory configurations.

B. Task Planning Across Deformation Manifolds

Task planning searches for a sequence of grasping and releasing actions that reaches the kinematic-equivalence class of the target under an appropriate correspondence. A releasing action is feasible only when the selected gripper is outermost within its grip and its release does not disconnect $\mathcal{G}_M$. Candidate grasping actions are generated from paths in $\mathcal{G}_M$. For each candidate, constrained optimization over the bending-angle vector searches for a self-collision-free endpoint at which the selected gripper overlaps its target grip with the prescribed grasping geometry. This endpoint test is substantially cheaper than full motion planning, but remains only a necessary feasibility check; continuous reachability is verified lazily after a candidate task plan has been found.

The continuous choice of grasping angles is discretized through correspondence construction. A constructive action uses a grasping angle appearing in the target configuration and thereby establishes a partial gripper correspondence. Auxiliary actions do not directly construct the target correspondence but may be required to make subsequent actions feasible. Together with releasing actions, these finite action classes permit search over the locally infinite graph.

For this search, we develop HEART-MCTS, a Monte Carlo tree search over a Hierarchical Expansion-As-Rollout Tree. Child nodes are first grouped by action type—constructive, auxiliary, or releasing—and constructive actions are further grouped by the target grasping angle they instantiate. This hierarchy allows exploration to be allocated both among action types and among alternative orders for constructing the target correspondence. Because conventional rollouts are expensive and no explicit simulation model is available, HEART-MCTS replaces rollout simulation with direct evaluation during expansion. Node values combine the available successor-action structure with progress toward reconstructing the target grasping angles. Previously visited angle systems are suppressed using an index-invariant encoding based on graph neural network, preventing redundant expansion. The search terminates after all target grasping angles have been instantiated, yielding a candidate sequence of inter-manifold transitions for subsequent motion verification.

C. Motion Planning Within Deformation Manifolds

Once a candidate task plan is obtained, motion planning computes the deforming actions required before each grasp and the final deformation required to match the target shape. Each deformation manifold is identified by enforcing angle-system consistency and self-collision avoidance. Angle-system consistency introduces loop-closure constraints for every independent cycle in $\mathcal{G}$. A minimum cycle basis is updated after each grasping or releasing action, and each cycle contributes three planar closure constraints on position and orientation. Polygonal approximations of module segments are used to enforce self-collision avoidance.

The resulting constraint equations and their Jacobians locally characterize the deformation manifold. For each selected transition, Atlas-RRT* computes a continuous path from the current configuration to the endpoint configuration

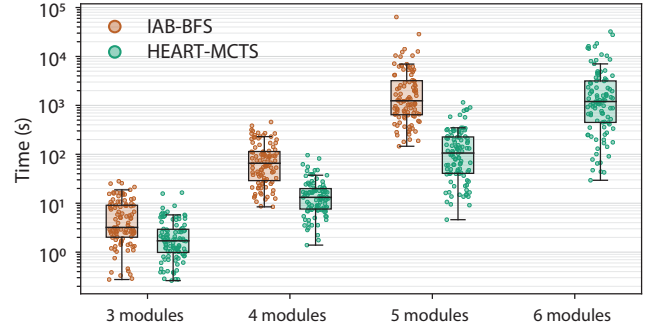


Fig. 4. Task-planning time for 100 procedurally generated start-goal pairs per module count. IAB-BFS was omitted for six modules because of its computational cost.

identified by the constrained feasibility check during task planning, while remaining on the manifold [10], [11]. As shown in Fig. 3B, different paths within the same manifold can reach grasp-preparatory states for different target grips, while varying the grasping angle along an overlap submanifold can induce transitions to infinitely many neighboring manifolds. Full motion planning is performed only after task planning returns a candidate action sequence.

IV. EXPERIMENTS AND RESULTS

To construct the evaluation set, MSRCR configurations were procedurally generated from discrete topology to continuous geometry. The gripper-level mixed graph $\mathcal{G}_M$ was first sampled using the configuration model, followed by cyclic-order assignment and the specification of bending and grasping angles. As a deterministic baseline, we also develop iterative auxiliary-budget breadth-first search (IAB-BFS), which uses the same correspondence-based action discretization as HEART-MCTS but limits the number of auxiliary actions along each search branch by a budget η_{AB} . Breadth-first search is repeated with increasing η_{AB} , analogous to iterative deepening. Because candidate grasp checks involve many independent constrained optimizations, these evaluations were parallelized across CPU cores.

Task planning was evaluated on 400 procedurally generated start-goal pairs with $N_m \in \{3, 4, 5, 6\}$, using 100 pairs per module count. As shown in Fig. 4, HEART-MCTS reduced average planning time relative to IAB-BFS by 59.5%, 82.0%, and 94.9% for three-, four-, and five-module systems, respectively; IAB-BFS was not evaluated for six modules because of its computational cost. Across the 300 cases with up to five modules, HEART-MCTS matched or improved the baseline grasp count in 97% of cases. Atlas-RRT* found initial feasible paths in 1.2 ± 0.7 s and 1.5 ± 1.2 s for five- and seven-module systems, respectively. Hardware executions further validated feasible transitions between robot morphologies.

Overall, these results show that the proposed geometric-topological representation supports efficient planning across discrete connectivity changes and continuous deformation manifolds. It provides a common basis for correspondence construction, task planning, and constrained motion planning in physically realized MSRCR reconfiguration.

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